

# From stay to play – A travel planning tool based on crowdsourcing user-generated contents



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## ABSTRACT

Conventional ways of trip planning using online reviews from multiple sources are often cumbersome and uncustomizable. The advance in information and communication technology (ICT) and the surge in user-generated contents (UGC) provide great opportunities to facilitate trip planning. This paper proposes a travel-planning tool by crowdsourcing multiple UGCs to provide customized information for tourists. We harvested hotel customer reviews from TripAdvisor, photo information from Flickr, and travel costs between destinations from Uber. First, we used geospatial data mining approaches to extract tourism attractions information from Flickr; identified multi-facet characteristics of hotels with natural language processing (NLP); and provided travel route recommendations with graph analysis. Second, we developed a web-based interface to let users communicate with the system interactively, which provides integrated recommendations including attractions, hotels, and visit route sequences. Two cities in the United States (i.e. Atlanta and Chicago) were used as cases to illustrate our approaches. The proposed travel planning tool is not only beneficial to support customized travel decision-making, but also supportive for hotel managers with strategic management implications.

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## 1. Introduction

The fast development of the information and communication technology (ICT) in recent decades makes information broadcast and sharing much faster and easier. The advancement in ICT and surge in user-generated contents (UGCs) have not only contributed to a new knowledge production mode through crowdsourcing (Wang, 2016), but also remarkably influenced many areas of human life, including the tourism industry and travel behaviors (Frew, 2000; Hjalager, 2015). According to a previous study, more than 87% of customers rely on online UGC to make purchase decisions for travels (Lipsman and Reston, 2007; Ghose, Ipeirotis, & Li, 2012). Travelers tend to seek advice from ratings and reviews prior to booking accommodations (Banerjee & Chua, 2016; Murphy, Chen, & Cossutta, 2016). For hotels managers, UGC can be used to predict different facets of business performance (Duverger, 2013), and web traffic volume to predict hotel demand (Yang, Pan, & Song,

2014).

Search engines have become one of the major avenues for travel planning and a critical component for tourism industry (Ghose et al., 2012). While a number of tourism websites provide information about destinations, accommodation and route selection, integrating and comparing different types of information from many contributors require significant time and effort. The number of choices makes it very difficult for consumers to find what they are looking for (Kabassi, 2010; Zhou, Xu, & Kimmons, 2015). Most tourists usually seek for information in three aspects before their travels: hotels to stay in, places to visit, and routes that connect hotels and attractions. There are difficulties in each of these search phases.

For seeking hotels, many websites only provide prices or star ratings as references. This approach reflects the overall or single product characteristics, but does not consider travelers' multidimensional preferences for hotels, such as location, service, room condition, facilities, and so on (Ghose et al., 2012). Websites such as TripAdvisor provide user generated textual descriptions. These descriptions are helpful in revealing details of hotels and providing multidimensional information. However, users have to go through

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a great number of reviews carefully to grasp the common themes and prominent characteristics of the hotel. Therefore, a tool that automatically summarizes a hotel's multidimensional attributes and qualities derived from diverse reviewers would greatly facilitate hotel search.

For searching tourism spots, websites apply different criteria when ranking popular destinations. The places with the highest rankings may not meet each traveler's needs and preferences. Hence, knowledge gained directly from actual travelers rather than lists of popular locations from websites can be more useful for tourists (Zhou et al., 2015). Information such as reviews, photos, ratings, and locations from previous travelers are valuable in planning trips. Increasing numbers of studies have used UGC to rank and recommend tourist spots based on spatial analysis and data mining techniques.

Apart from the tourism spots and hotel choices, routing is another important aspect in travel planning (Sun, Fan, Bakillah, & Zipf, 2015). Most current travel recommendation systems focus on tourist spots and hotels; few of them provide routing service. Some platforms may have provided the route choices for tourists, but these routes are usually predetermined by travel agencies and are not flexible enough for individual travelers (Liu, 2006). Others may rely on the shortest route connecting destinations, although the shortest paths are not always the best route plans (Popescu & Grefenstette, 2009). Factors such as price and relative location between hotels and attractions also play important roles. It would be valuable to design a routing service considering factors including distance, price, and relative location between hotels and attractions for tourists.

This paper aims to develop a geospatial analytical tool to assist decision-making in travel planning. This tool provides integrated services for hotels, attractions, and routing based on crowdsourcing data from social media, travel website, and Uber application programming interface (API). The inputs of the system are reviews of hotels, Flickr images, and Uber taxi costs between places. The outputs of the tool are the recommended attractions, hotels, and visit route sequences. We use geospatial data mining to extract attractions from Flickr images, identify multidimensional characteristics of hotels using natural language processing (NLP), and suggest travel routes and sequences using graph analysis. Two cities in the United States (i.e. Atlanta and Chicago) were used as case studies to illustrate our approaches. We then developed a web portal to deliver the recommendation service. In the following section, we introduce the related literature and identify the gaps. Section 3 presents the proposed methods geospatial data mining. Section 4 describes results for each component of the systems using data from Atlanta and Chicago. Finally, Section 5 provides a general discussion of the tool and its contributions.

## 2. Related work

### 2.1. Extracting hotel information based on UGC

Traditionally, information regarding qualities of hotels are generally represented through the stars and price. Empirical evidence shows that price has a positive impact on perceived quality (Ye, Li, Wang, & Law, 2014), and the official rating (the number of stars) of a hotel is a good indicator of quality (Núñez-Serrano, Turrión, & Velázquez, 2014). With a proliferation of the UGC on hotel reviews, an increasing amount of studies start analyzing and recommending hotels based on suggestions left by previous customers (Ghose et al., 2012). Most studies only focus on numeric review ratings (e.g. rating scores and/or review volume), and as such, they may only reveal one dimension of hotel characteristics. They also suffer from deviance from model randomness because

people are more motivated to leave reviews when they had extremely good or bad experience (O'Connor, 2008). Similarly, reviewers may only rate the hotels they stay repeatedly, leading to that fact that the average rating score for a hotel may not be the population's average rating (Ghose et al., 2012). To mitigate above-mentioned limitations, this study computes hotel rating scores for multiple facets from actual review contents through NLP in conjunction with text mining and sentiment analysis.

Another strain of studies use text mining and sentiment analysis to analyze the textual reviews (Das & Chen, 2007; Ho, Lieberman, Wang, Samet, & A. C. M. and Sigspatial, 2012; Hu & Liu, 2004; Kasper & Vela, 2011; Stepchenkova, Kirilenko, & Morrison, 2009; Xiang, Gretzel, & Fesenmaier, 2009). Sentiment analysis has been widely used to reveal people's feelings about certain targets based on textual expression. Such information has many applications in customer review analysis and recommendation systems (Yang, Zhang, Yu, & Wang, 2013). With the increasing amount of social media data, sentiment has been used to detect the temporal mood pulse towards political election (Diakopoulos & Shamma, 2010), quality of geotagged services (Cheng, Caverlee, Lee, & Sui, 2011), geography of happiness (Mitchell, Frank, Harris, Dodds, & Danforth, 2013), and non-personal tweets (Guo & Chen, 2014). Most sentiment analyses are used to evaluate the overall positive or negative perceptions of a particular topic or service. When the piece of text is short, such as for tweets (maximum 140 characters), it may just contain a homogeneous sentiment (positive or negative). However, for longer reviews, mixed sentiments may be embedded in the text, resulting in multidimensional information. Recent studies have started to investigate the sentiments for specific hotel characteristics (Cataldi, Ballatore, Tiddi, & Aufaure, 2013; Matsuyoshi, 2015), but most current recommendation systems have not considered these factors in assisting travel planning. Therefore, this study takes one step further in extracting multidimensional information from reviews and applying sentiment analysis to each dimension, hence assigning quantitative values to multiple dimensions to assess.

### 2.2. Tourism spots detection

The rapid increase in UGC has attracted growing research interests in tourism spots detection using social media data. For location based analysis, spatial clustering methods, such as K-means (Kennedy & Naaman, 2008), and mean shift (Yin, Cao, Han, Luo, & Huang, 2011, pp. 980–991), DBSCAN (Hu et al., 2015; Sun et al., 2015; Zhou et al., 2015), or its modified version for photos, *p*-DBSCAN (Kisilevich, Mansmann, & Keim, 2010), have been used in many previous studies. Clustering methods have also been applied in mining features for images. Kennedy and Naaman (2008) used global and local features extracted as photo signatures to group similar images. The Scale-Invariant Feature Transform (SIFT), which detects signatures that are irrelevant to the scale and rotation of images have been widely used (Lowe, 2004). For semantic tags, different algorithms have been used to process the unstructured textual information. Ahern, Naaman, Nair, and Yang (2007) analyzed the geo-referenced tags from Flickr images to generate aggregated knowledge for arbitrary areas in the world. A visualization tool, World Explorer, was also created to illustrate the extracted information. In this study, we used tourism spots detected in our previous work which used the locations, tags, and temporal characteristics of Flickr images as the basis for the recommendations (Zhou et al., 2015).

### 2.3. Routing

Travel route planning is another important step for tourists. Previous studies have used road network analysis, impedance

model, and shortest distance to conduct route planning. Sadeghi-Niaraki, Varshosaz, Kim, and Jung (2011) used a geographic information system (GIS) based tool to plan the tour routes considering weather, sightseeing information, and road types using an impedance model. Shih (2006) investigated the road network characteristics using graph indices and recommended the appropriate tourist services of each particular destination based on the structural characteristics of network position on various touring routes. Zhan and Noon (1998) summarized main shortest path algorithms based on real road networks. In addition to road network-based approaches, recent studies used social media data to generate and recommend trip routing. Based on the information discovered from geo-tagged photos, Lu, Wang, Yang, Pang, and Zhang (2010) proposed algorithms to generate popular destinations and the visiting order of the destinations. Vu, Li, Law, and Ye (2015) used Markov Chain to mine travel patterns based on Flickr data, presenting the travel sequence preferences for Asian and western tourists. Yin et al. (2011) first discovered the trajectories from travelers and then developed an exemplar-based algorithm to rank and recommend the paths. In addition to using the social media data, GPS records have been used to recommend users with top interesting locations and travel sequences (Zheng & Xie, 2011). Overall, different factors such as distances, scenes, and popularity in route selection have been considered in previous studies. Relatively little attention has been paid to the cost of taxi service. This study investigates tour route planning to minimize the cost of Uber rides. For route planning, algorithms such as traveling salesman problem (TSP) have been proposed to search for optimal routes when one travel through each destination only once before returning to the origin. However, TSP usually considers homogeneous destination types, such as only attractions. This study extends such methods to take both attractions and hotels into consideration. The goal is to give recommendations so people can depart from and arrive at an ideal hotel among all hotels and visit attractions-of-interest exactly once at a low cost.

### 3. Data and method

Data used in this study were collected from multiple sources of UGCs. We used worldwide Flickr dataset, which is obtained from Flickr Creative Dataset from Yahoo Lab. This data contained 99.3 million images, 49 million of which were geo-tagged. The hotel review data was collected from TripAdvisor, one of the most popular sources of hotel ratings with more than five million registered users (Banerjee & Chua, 2016). For travel routes, we collected Uber minimum and maximum prices and distances between hotels and tourism attractions using Uber API. In the following section, we present the methods in identifying both hotels and attractions as well as suggesting optimum route sequences that connect hotels and attractions.

#### 3.1. Extracting multidimensional hotel information from reviews in TripAdvisor

We first aggregate all reviews for one hotel to a text corpus. All reviews that are not written in English were removed from the dataset. We used Stanford CoreNLP to process the textual data (Manning et al., 2014). Sentences in each review paragraph were separated. Each sentence was sent to a dependency analysis module to parse the structure of the sentence.

We used the dependency structure to parse sentences. Dependency in NLP is a binary asymmetric definition that describes how words relate to other words (Kao & Poteet, 2007). A dependency relationship contains a governor word and a dependent word. For instance, the expression “clean carpet” consists of a

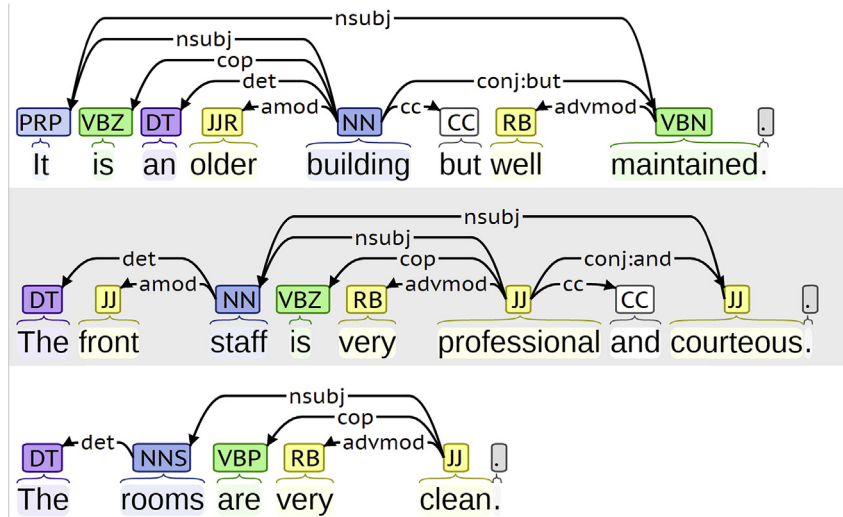
“carpet” as the governor word and “clean” as a dependent. Each sentence is presented by a labeled directed graph  $G = \{N, E, L\}$ , where  $N$  is a set of words as nodes,  $E$  represents edges as the relationship between a pair of words, and  $L$  represents the types of dependency for each edge. For instance, in the sentence “the room is cozy”, there is a nominal subject relationship between room and cozy *nsubj(cozy, room)*. There are about 50 dependency types in CoreNLP libraries (De Marneffe et al., 2008). Fig. 1 provides a dependency structure for one sample review: “It is an older building but well maintained. The front staff is very professional and courteous. The rooms are very clean”. The dependency structure works well for most standard sentences, but sentences that are grammatically incorrect may affect the accuracy.

The process to transfer dependency graphs of each review to a quantifiable representation involved two steps. We first generated all dependency graphs for reviews. Then, three types of dependencies including nominal subject (*nsubj*), adjectival complement (*acom*), and negation modifier (*neg*) were extracted for future analysis. Sentences with *neg* relationship were discarded from the analysis, because they usually contained the opposite meaning of the literal word. Second, we searched for part-of-speech (POS) tagging in the dependency graph edge. POS tagging is a process used to analyze the grammatical parts of the word based on words' definitions and contexts. In this study, we mainly focused on nouns and adjectives. For dependency edges, we created a hash table  $H = \langle \text{noun, descriptions} \rangle$  to represent the hotel attributes. For each attribute, we created a sub hash table  $D = \langle \text{adjective, \# of occurrence} \rangle$  to retain information regarding how the particular attribute is being described. Each hotel thus was represented by a large hash table  $H$  that contained different features of the hotel.

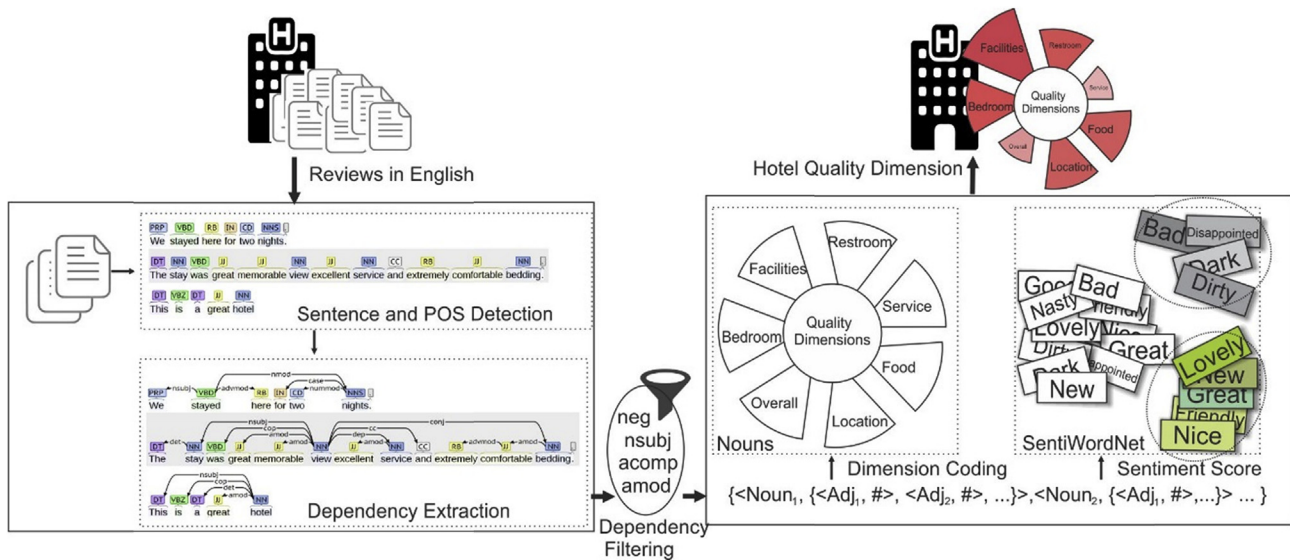
After obtaining the  $H$  tables, we summarized the attribute dimensions for hotels. Adapted from the Criteria for Hotel Operators and Quality Grading (Northern Ireland Tourist Board, 1982), we summarized the qualities of hotels into seven dimensions: public area and facilities, bedroom condition, restroom condition, service, restaurant and food, location, overall condition (Fig. 2). We manually coded the derived nouns in the hash tables for all hotels into appropriate classes. Table 1 shows the example nouns for each category.

We used the SentiWordNet 3.0 English lexical resources to infer the sentiment of adjectives that modify the nouns representing qualities of hotels. SentiWordNet is publicly available for supporting sentiment classification and opinion mining applications (Baccianella, Esuli, & Sebastiani, 2010). The background database WordNet includes a rich set of nouns, verbs, adjectives, and adverbs in different cognitive concepts and sentiment scores. We calculated the sentiment score for each adjective in hash table  $D$ . We then created a weighted sum by multiplying the scores of each adjective by the number of its occurrences to represent the quality of the corresponding hotel attribute. Because each selected attribute (noun) was associated with one of the seven dimension of the hotel qualities, we aggregated the scores for each category to generate an overall evaluation for each attribute dimension for each hotel.

For instance, according to our methods, the following review message “In general, XYZ hotel is not bad. The staff are very welcoming and friendly. Great food, the yogurt is absolutely yummy!!! However, it smells moldy in the room. The bed is uncomfortable ...” showed that the score for Restaurant and Food was 0.612 while the score for Room condition was  $-0.523$ . This represented only one review for one hotel. We summarized all reviews for this hotel to generate the overall scores for the seven dimensions. The proposed method can clearly reflect the dimensionality and polarity of the hotel quality.



**Fig. 1.** Basic dependencies for the sample sentence. Color labels represent part-of-speech (e.g. NN stands for noun, JJ stands for adjective, VBD stands for verbs in past tense). Arrows connecting words represent grammatical dependencies (e.g. amod represents an adjectival modifier, advmod represents an adverb modifier). The texts were from actual reviewer comments and figure was generated based on the Stanford CoreNLP service (<http://nlp.stanford.edu:8080/corenlp/process>).



**Fig. 2.** The process to obtain the dimensions of hotel quality.

**Table 1**

Example nouns that describe different aspects of hotels.

| Quality dimensions         | Representative nouns                                             |
|----------------------------|------------------------------------------------------------------|
| Public area and facilities | parking, pool, lobby, amenity, elevator, lounge, store, hallway  |
| Bedroom condition          | bed, pillow, carpet, sleep, bedding, sheet, closet, couch, A/C   |
| Restroom condition         | tub, bath, bathroom, shower, water, towel, toilet, soap, shampoo |
| Service                    | reception, bartender, cleaning, convenience, massage             |
| Restaurant and food        | drink, meal, fruit, cookie, cocktail, waffle, lunch, seafood     |
| Location                   | location, distance, proximity, neighborhood, street              |
| Overall condition          | place, experience, review, quality, condition, impression        |

### 3.2. Hotspot detection

We began the tourism destination search process by collecting geo-tagged image for a given city. We mainly used three types of information from Flickr to detect the attractions, including the textual tags, the geographic locations, and the timestamps. For tag

attributes, we focused on importance and relevance of tourist destinations as the two main considerations. We calculated five indices, including number of occurrence times for each tag (Ntag), time expansion of each tag (Ttag), area of the bounding box of each tag (Atag), number of users associated with each tag (Utag), and entropies for three temporal scales (Etag).



Ntag indicates how popular each tag is. Greater Ntag value in general is associated with higher popularity. Ttag is mainly used to differentiate tourism spots from events. Tourism spots tend to have many photos year round while event-related photos only occur in a short period of time. Atag represents how spread a tag is. Tags related to tourism destinations usually concentrate in relatively small areas. Utag indicates how popular a tag is among different Flickr users. Etag measures the degree of randomness of travel patterns, which is also a useful feature for separating tourism spots from event-related photos.

These tag features are then sent to a supervised classification model to rank features based on their relevancy to the tourism spots. We also conducted a clustering method based on a tag-similarity graph to group related tags together. For location characteristics, we used a DBSCAN algorithm to identify potential locations for tourist spots. To refine the clusters, we used both geographic features and tag semantics to aggregate small clusters together using a rule-based algorithm. We also developed an algorithm to rank the identified tourist spots based on their popularity. The details of the algorithm is described in our previous work (Zhou et al., 2015).

### 3.3. Routes connecting attractions and hotels

Route sequences may be determined based on many factors, such as cost, aesthetics, distance, and safety. To simplify the model, we used average Uber cost to recommend travel sequences that connect attractions and hotels. The task to find an ideal route sequences connecting attractions and a hotel is similar to the traveling salesperson problem (TSP) algorithm. The core of this question would be, given a set of hotels and attractions, what is the shortest possible route that starts from one hotel, visits each attraction only once and returns to the hotel. To solve this problem, for each hotel, we need to create length of  $n!$  permutation vectors to compare the shortest cost path, where  $n$  is the number of attractions. This kind of brute force approach requires a factorial running time. When the number of planned visiting attractions increase, the computational load increases significantly.

Hence, rather than using the permutation approach, we used heuristic algorithms to solve this question. Although it may not be a global optimal solution, modern heuristic algorithms can reach 2–3% away from the optimal solution with a fast running time. The distance away from the optimal solution is enough for a travel route planning application (Rego, Gamboa, Glover, & Osterman, 2011). In this study, we used a modified arbitrary insertion algorithm with 2-Opt refinement to solve the problem. All attractions to be visited were put in a complete graph  $G = (N, E)$ , where attractions were listed in the node set  $N = \{1, 2, 3 \dots n\}$  and each edge  $e_i \in E$  contained a distance  $d_i$  between the attractions it connects. We pre-computed the Uber cost matrices between hotel-spot distance  $M_{hsdist}$  and price cost  $M_{hscost}$ , spot-hotel distance  $M_{shdist}$  and price cost  $M_{shcost}$ , and spot-spot distance  $M_{ssdist}$  and price cost  $M_{sscost}$ . Not all routes that connect two spots need an Uber ride. Some tourists prefer walking between places, especially if the distance in between is short. However, tourists may have different preference in how long they would like to walk. To factor in this variability, we implemented a slide bar in the web interface that allowed users to adjust the distance threshold above which they would prefer a ride to walking. If the distance between two sites was shorter than the threshold, there would be no cost involved in transportation. Otherwise, the average cost of Uber between such two sites would be used as model weight.

To start with, we used the arbitrary insertion algorithm to create a tour sequence. To do this, We first randomly selected a destination  $i$ . Then we identified the second destination  $j$  for which  $C_{ij}$  (i.e. cost

from  $i$  to  $j$ ) is minimum and build the partial sequence  $(i, j)$ . Second, we arbitrarily selected a third destination  $k$  that was not yet in the partial sequence, and inserted  $k$  into the existing sequence between the two consecutive destinations  $i$  and  $j$ , in the way that minimize the cost  $C(i, k) + C(k, j) - C(i, j)$ . This process continues until all destinations are inserted. This is the process to construct the tour sequence.

Then, we used the 2-Opt refinement approach to improve the model. Basically, we iteratively removed two edges and replaced them with two different edges that reconnected the fragments to a new graph with a shorter total distance cost. This refinement procedure continued until no further improvements were possible. Once the route sequence  $S$  with the shortest distance was created, the pairwise list of adjacent nodes from  $S$  were retrieved to compute and suggest  $h_i \in H$  hotels. For each sequence that connected a pair of attractions  $N_j$  and  $N_k$ , a hotel node  $H_i$  was inserted to connect attractions  $N_j$  and  $N_k$ . The distance and cost increase due to the insertion of the hotel were recorded. The system iterated through all the possible insertions and calculated the five lowest cost after adding the hotel nodes (Algorithm-1). In other words, the edge  $e_{(k,j)} \in E$  that connects to attractions  $N_j$  and  $N_k$  was replaced by the routes connecting to the Hotel  $i$ :  $e(N_j, H_i)$  and  $e(H_i, N_k)$ . This algorithm can be easily extended to situations with multiple sequences, which is described below.

In fact, not all tourists would loop through all the tourism destinations in a day-trip. A multi-day trip would be more common for those who wish to visit a lot of destinations in a city. One solution to model this behavior would be manually splitting tourism spots and recommending a subset of destinations for each day. However, it is difficult for users to decide how to split the destinations, especially when they do not know the location for each spot. Furthermore, choosing a hotel that minimizes the transportation cost for all travel paths would be challenging. To solve this problem, we proposed two steps to find a suitable travel plan for multiple days and ideal hotels to stay.

We created a web-interface to obtain user preference regarding trip duration and display the recommended routes calculated based on the user's input. We first split the spots planned to be visited into multiple clusters based on the number of days of travel, which is entered by the user. To reduce the travel distances and cost for any given day, we grouped destinations that were closer to each other into clusters using k-means clustering. The number of

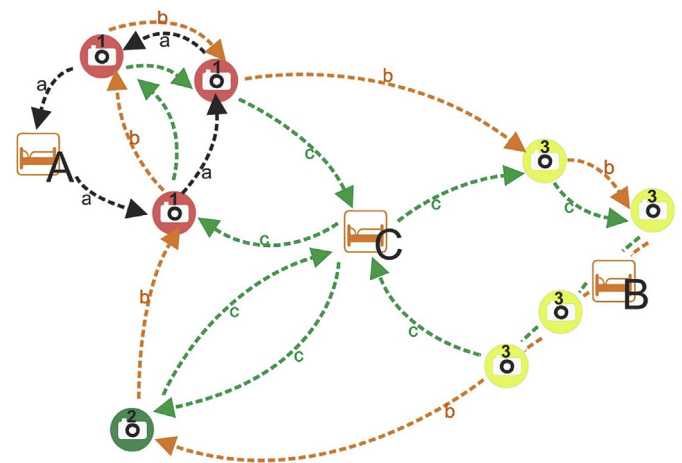


Fig. 3. Demonstration of the method to heuristically minimize travel cost for three clusters. Hotel A and Route a will minimize the cost for cluster 1 (3 spots). Hotel B and Route b will minimize the cost for all spots (8 spots). Hotel C and Route c will minimize the cost for three clusters on three days (8 spots).

clusters was determined by the number of days provided by the user. Each cluster represented a number of destinations that could be visited in a single day. After that, we then identified the best hotel and the lowest cost route on top of Algorithm -1. Fig. 3 shows the concept. Among all the filtered hotels based on the seven dimensions, Hotel A and route a will be a low cost solution for Cluster 1 (or Day 1) based on Algorithm-1. However, this hotel will be only good for Day 1. Hotel B and route b will be good if the traveler wants to visit all spots in one day. In contrast to the previous two scenarios, we added up the computed cost for each cluster and selected the least five options as the best solutions. Hotel C and route c is an example that generates a low cost considering all clusters.

## 4. Results

### 4.1. Hotel multi-dimensional features and selection

To demonstrate this approach, we tested the system using data from Atlanta and Chicago. We collected 175 hotels in Atlanta and 162 hotels in Chicago. To adjust for the number of reviews, we normalized the review scores by the number of reviews for each hotel. Table 2 represents the descriptive statistics for the seven dimensions that represent the quality of hotel in both cities. Overall, the distribution of the scores reveals similar patterns in the two cities. Service quality received the highest scores and bedroom condition has the lowest scores in both cities.

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#### Algorithm-1

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**Input:**  $M = \{M_{hscost}, M_{hsdist}, M_{shcost}, M_{shdist}, M_{sscost}, M_{ssdist}\}$  Matrices for Hotel-Spot distance and price cost, Spot-Hotel distance and price cost, and Spot-Spot distance and price cost;  $H_{ids} = \langle h_1, h_2, h_3, \dots, h_n \rangle$ : potential hotel ID list;  $S_{ids} = \langle s_1, s_2, s_3, \dots, s_m \rangle$ : potential spot ID list.

**Output:**  $RS = \langle r_1', r_2', r_3', \dots, r_{m+1}' \rangle$ : the route sequences that connect each spots and a recommended hotel

**Construct**  $G = (N_m, E_m)$  where  $N$  is the set of  $m$  spots planned to be visited,  $E$  is the cost edge between pairwise nodes.

**Create** a route sequence  $rs = \langle r_1, r_2, r_3, \dots, r_m \rangle$  based on the arbitrary insertion approach

$d = \text{CalcTotalCost } rs$

**for** all possible **in** edge-pairs

$rs' = \text{Swap existing } rs$

$d' = \text{CalcTotalCost } rs'$

**if**  $d' < d$

**Update**  $d$  &  $rs$

**return**  $rs = \langle r_1', r_2', r_3', \dots, r_m' \rangle$

$hCost = 999$  /\* initial trip to hotel cost \*/

**for each**  $r_i$  **in**  $rs$

**for each**  $h_j$  **in**  $H_{ids}$

$hCost' = \text{CalcCost } c_{ris-hj} \ \& \ c_{hj-rie}$  /\* The cost between route  $r_i$  start point  $s$  to hotel  $j$  and the cost between hotel  $j$  to  $r_i$  end point  $e$  \*/

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**if**  $hCost' < hCost$

$hCost = hCost'$

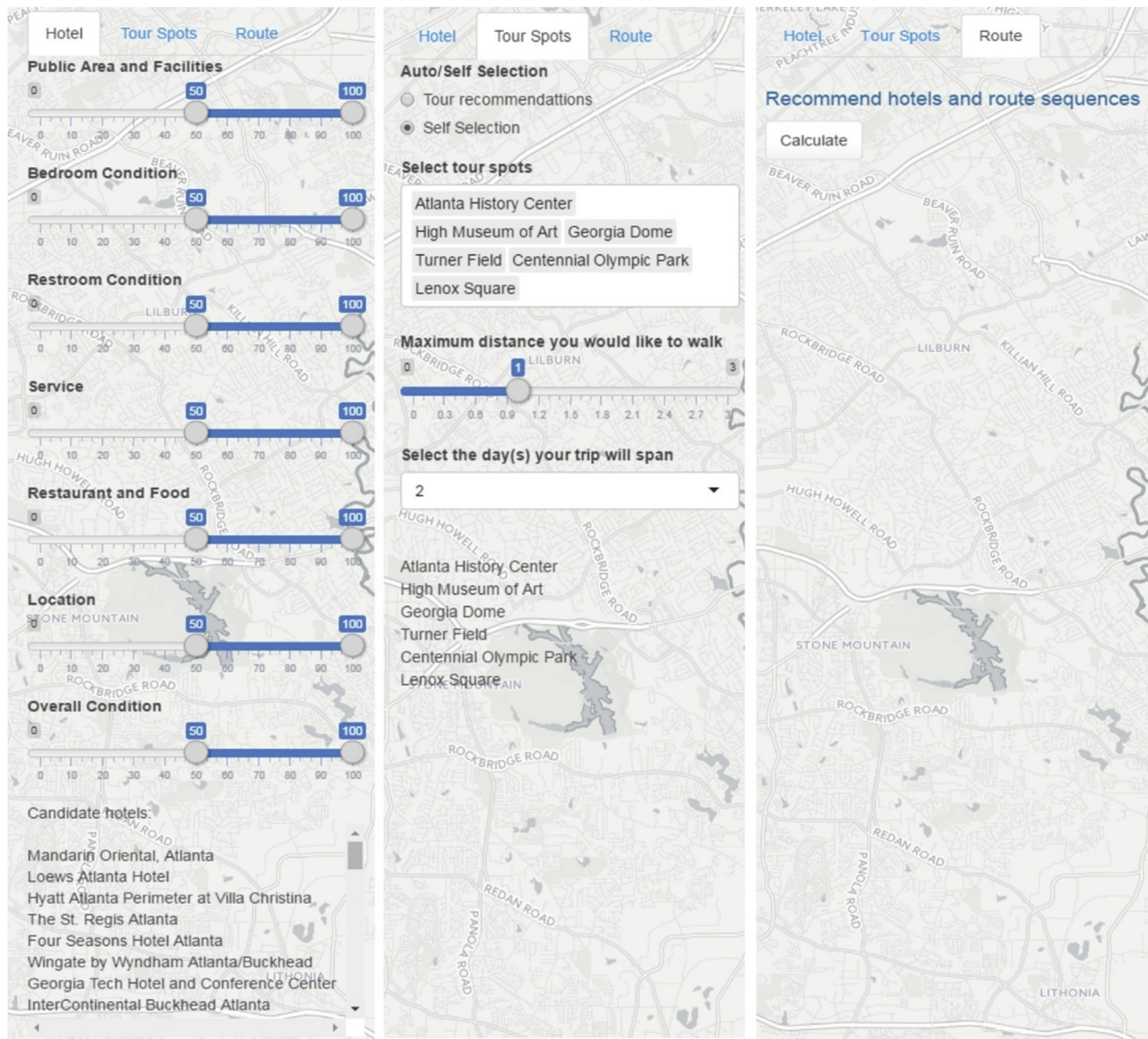
**return**  $RS = \langle r_1', r_2', r_3', \dots, r_{m+1}' \rangle$  /\* replace one  $r_i$  with the detected two routes (the shortest route from and to hotels) \*/

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### 3.4. Web application development

We developed a web interface using Shiny and Leaflet for R to provide the recommendation service to the public. Shiny is a web application frame for R, which can quickly turn the analysis components into a web application service (Shiny, 2015). Fig. 4 shows the web-interface for the application based on the proposed methods. The interface provides three panels: Hotel, Tour Spots, and Routes. The Hotel panel provides slide bars for users to select hotel preferences in the seven dimensions of hotel qualities. The Tour Spots panel provides options for users to choose either the top N spots to visit or manually select the ones they are interested in. Lastly, the Routes panel shows the recommended routes and the estimated Uber Taxi cost.

We also created a quality index by adding up scores for each dimension (*Sum*). There is a general consistency between the average rating score from generic numeric rating (*Score*) and the quality index we proposed through NLP (*Sum*). A robust bivariate linear regression shows the proposed quality index (*Sum*) itself can explain as much as 80% of the total variance of the generic numeric rating (*Score*). Fig. 5 provides a scatter plot for the generic number rating and the quality index. The two outliers in the lower left corner of Fig. 5 are two hotels in Chicago, where there is only one review for each hotel. Additionally, we have also confirmed that the proposed quality index through NLP (*Sum*) is consistent with the hotel rank (*Rank*) at both cities (Fig. 6). Specifically, when regressing *Sum* on *Rank*, the  $R^2$  reaches 0.70 for Atlanta and 0.88 for Chicago (robustness bivariate linear regression). Again, the two outliers in the lower right corner of Fig. 6 are those two hotels with single review in Chicago.



**Fig. 4.** The web interface for the proposed recommendation system. On the Hotel panel, when adjusting the threshold of each dimension, the candidate hotels will be updated accordingly in the box below. On the Tour Spots panel, a user can choose either top N spots to visit or manually select the ones they are interested in. The results will be updated accordingly in the box below. The threshold of Uber ride and trip duration can be adjusted as well. On the Route panel, once the “Calculate” button is clicked, the system will compute and recommend routes connecting hotels and spots.

**Table 2**

Distribution of the quality score regarding the seven dimensions for the hotels in two cities.

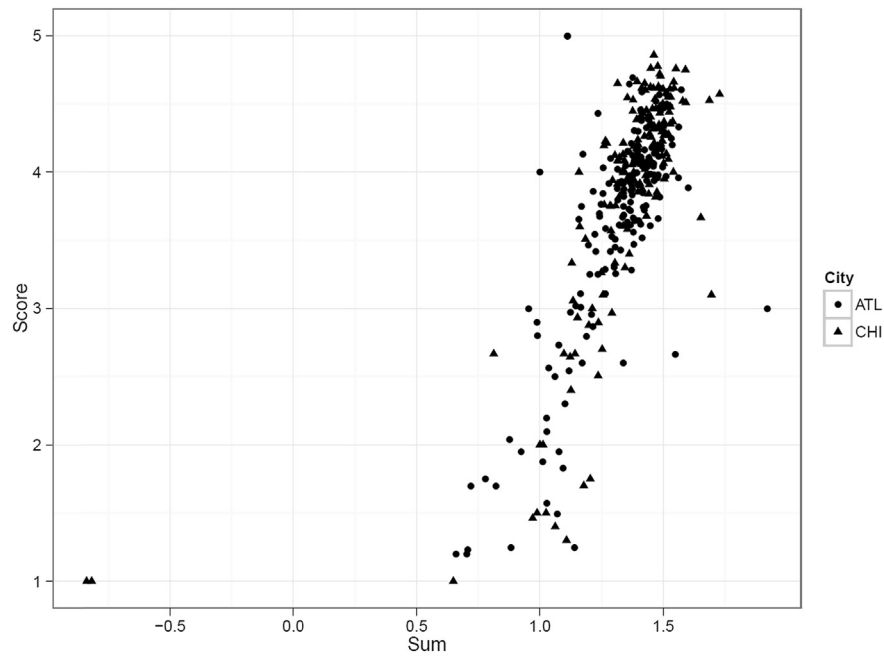
| Quality dimensions         | Atlanta scores |            | Chicago scores |            |
|----------------------------|----------------|------------|----------------|------------|
|                            | Mean × 100     | s.d. × 100 | Mean × 100     | s.d. × 100 |
| Public area and facilities | 2.12           | 3.63       | 3.27           | 2.88       |
| Bedroom condition          | 0.10           | 1.12       | 0.76           | 1.17       |
| Restroom condition         | 4.28           | 5.99       | 4.95           | 8.08       |
| Service quality            | 12.20          | 7.29       | 9.42           | 19.21      |
| Restaurant and food        | 6.72           | 5.51       | 6.26           | 5.60       |
| Location                   | 4.72           | 3.46       | 8.18           | 4.82       |
| Overall condition          | 12.85          | 12.65      | 15.79          | 18.57      |

Then, the quality in each dimension was mapped based on the scores from 0 to 100. Users can specify the desired hotel quality threshold in each dimension. This threshold will serve as the

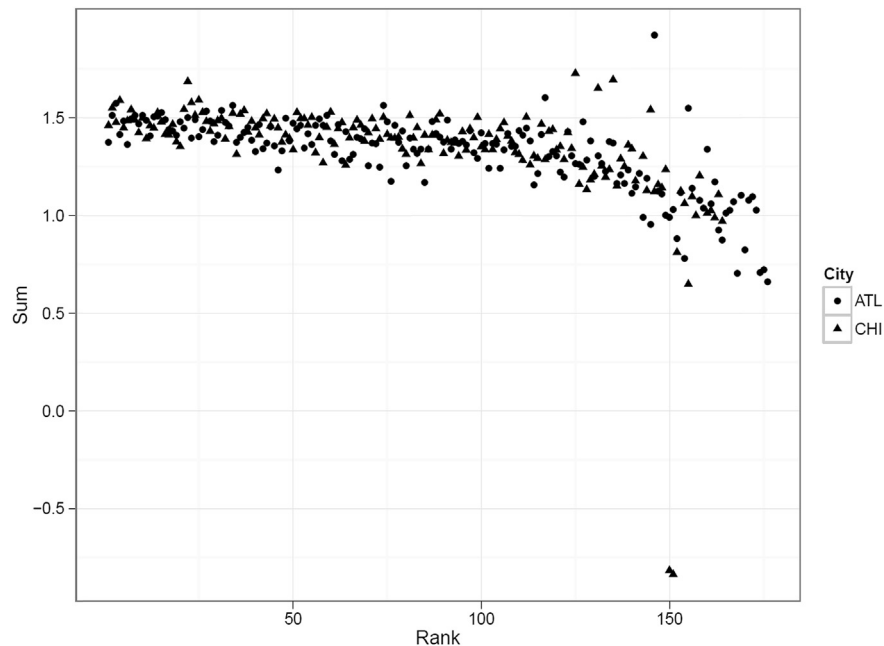
quantile cutoffs for each category to subset hotels. In other words, the system will generate a set of hotels for each dimension, and the hotels that meet the threshold for all seven dimensions will be selected and displayed for the final hotel candidates. For instance, if we want to select a hotel with 80 or above scores on all quality dimensions, we will obtain nine hotels in Atlanta and ten hotels in Chicago.

#### 4.2. Tourism spot selection

For each city, the developed algorithm generates ranks for each tourism spot. Users will have the options to select the top N spots they want to visit, or arbitrarily select the spots they are interested in from the given list.



**Fig. 5.** Scatterplot showing the correlation between our proposed quality index score (*Sum*) with average generic scores (*Score*). ATL = Atlanta, CHI = Chicago.



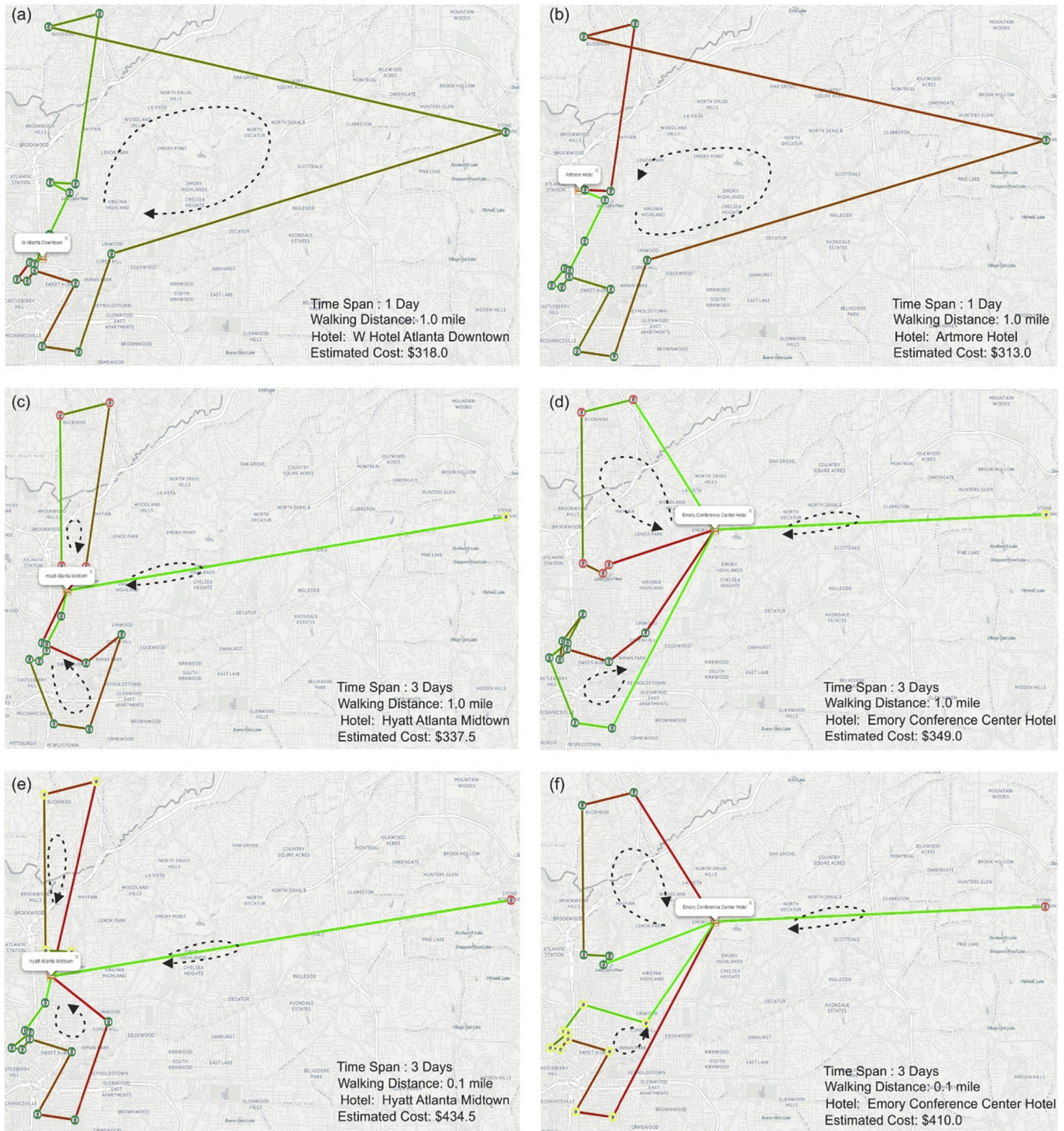
**Fig. 6.** Scatterplot showing the correlation between our proposed quality index score (*Sum*) with hotel rank (*Rank*). ATL = Atlanta, CHI = Chicago.

#### 4.3. Route suggestions

Based on the given hotel and attraction sets, the proposed method will run to find the preferable route sequences. Fig. 7 shows the result for three scenarios. Fig. 7a and b show the results for Scenario 1: All trips complete in 1 Day. We showed options for two hotels. The total costs are similar in both cases. In Scenario 2, the user plans to spend three days to tour around the city. The system divides all spots into three clusters based on spatial closeness. We can see major differences between route sequences between Scenario 1 and 2. However, the difference between transportation

costs between these two scenarios are not as salient. The main reasons for this is because the Stone Mountain (the third cluster) is far away from the first two clusters. The cost between the hotel and the Stone Mountain accounts for a major portion of the total cost, which is shared by the two scenarios. A small cost increase comparing to Scenario 1 is because of the round trips between hotels and attractions in Day 1&2 (Fig. 7c and d). In Scenario 3, the user enters a shorter Uber ride threshold since he prefers to walk less (0.1 mile comparing to 1 mile in previous two scenarios). From the results of this scenario, there are some minor changes in route selection, while the transportation cost is much higher than the





**Fig. 7.** Routing sequences (dashed lines) that connect hotels and the specified tourism spots. Scenario 1: One day trip (a&b). Two different hotel recommendations and route sequences are recommended. Scenario 2: Three day trip (c&d). The route sequences are very different from Scenario 1. Scenario 3: Prefer less walking (e&f). This option results in much higher cost in transportation compared to Scenario 2.

previous two. In this case, the difference in cost reflects the Uber ride threshold more than the trip duration.

## 5. Discussion and conclusion

In the era that social media and UGC become increasingly ubiquitous, more travelers tend to rely on a large number of ratings

and reviews prior to making a travel purchase. Accommodation is one of the biggest components of a travel budget; therefore, consumers are more likely to consult with a large variety of Internet-based sources prior to making the actual purchase. TripAdvisor, as an early adopter of UGC, is the world's largest third-party travel site. A concern of UGCs is that the information quality and reliability. To address such issue, TripAdvisor enables its quality check



protocols to maintain its relative high quality of contents. For example, it scrutinizes IP address and email address of the UGC contributors, and tries to detect any suspicious patterns or obscene or abusive language before posting any review to the website instantly. Such strategies largely ensure the quality of its UGCs in evaluating hotel qualities.

It is promising to use NLP to extract sentence components that describe certain aspects of hotel qualities. Studies using sentiment analysis to detect hotel qualities based on entire sentences may neglect important features. For instance, sentences contain positive comments on food and negative comments on rooms may be canceled off and generate a neutral sentiment. One contribution of this method is to polarize different facets of hotel attributes using dependency graphs including: nominal subject (nsubj), adjectival complement (acomp), and negation modifier (neg). Qualities of hotels reflected by reviews can be clearly reflected. This practice has largely enriched information available to both consumers for their purchasing behaviors and hotel managers for their management strategies. For travelers, categorizing UGC of a hotel into seven dimensions allows users to select a hotel that meets their individual needs and personal preference about particular items. Hence, it enables a more customized hotel selection for everyone. For managers, customer satisfaction in multiple dimension would help evaluate hotel qualities comparing with other hotels in the market.

However, there are several limitations of this study related to the application of NLP. First, we cannot extract quality-related information from all relevant sentences. For example, sentences such as “Will definitely stay again” or “Noise made me sick” will not be captured despite providing meaningful information in quality evaluation. In addition, the number of comments regarding the seven dimensions varies a lot. For instance, comments about services are much more than locations. For hotels with relatively low volume of comments, it is difficult to provide reliable quality assessment in all dimensions. Future work may integrate other UGC such as Twitter or Google Reviews as well as geographic information such as OpenStreetMap data to better assess hotel attributes. Additionally, exogenous factors such as proximity to high-quality restaurants, neighborhood safety and environmental aesthetics may affect a traveler's hotel selection behavior. Another direction of future work is to incorporate such restaurant quality information (e.g. from Foursquare, Yelp, etc.), socioeconomic profiles (e.g. from American Community Survey) and environmental data (e.g. from remotely sensed imagery) to provide comprehensive hotel recommendation based on a traveler's needs.

Regarding the route sequence, we modeled the sequence to minimize the Uber cost. Uber gains increasing popularity among travelers in recent years. For travelers who are not familiar with local transportation, Uber turns to be an ideal means of transportation. Another useful feature for Uber dataset is that the distance calculated between origin and destinations are based on actual road networks, which represent the transportation demand realistically. For instance, distance and cost from point going from A to point B may be different from those from B to A. Uber provides minimum and maximum costs between origin and destination. In this study, we used the mean cost between the minimum and maximum as a benchmark. The contribution of the route planning resides in quickly finding good sequences that link hotel and tour spots and reduce the cost. Although the sequence may not be the global optimal solution, results should be at most 2–3% away from the optimal solution (Rego et al., 2011). When planning the route, this study, however, did not consider the time user spend on visiting each spot. One future solution to this issue is to estimate a duration of stay for each spot by mining UGC. For instance, stay time might be extracted from Flickr photo trajectory based on

individuals who upload the photos while they are traveling. However, we have not explored the feasibility in this study.

Last but not the least, an assessment of the tool is necessary before it is made available as an application to tourists and managers. We plan to do pilot tests to evaluate the system in multiple cities among different population groups regarding the usefulness and feasibility of the tool.

The main goal of this project is to examine if we can leverage UGCs from multiple sources to provide recommendations at the different stages of tour planning. We believe our results suggest that information gained through mining the UGCs will provide valuable references if not solutions to tourism planning.

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